M5A-R3 PRIVATE DL3BBX REFERENCE DERIVATIVE / NOT RELEASED
Original72-to144MHz doubler,2N4427 driver,MRF237 final and matching retained.
Si5351 CLK0=RF/2; CLK2=RF. No premium MMIC and no speculative LDMOS matching.
BLOCKING: exact RF transistor procurement/manufacturer and some package/magnetic geometry.
Published >3WPEP is not proof of requested5.135WPEP connector capability.
All RF/AM/thermal/endurance hardware tests NOT_RUN. No PCB or procurement authorized.
Private construction permission only; commercial derivatives require author written permission.

01 Interface and local SI5351 source

File: 01_interface_source.kicad_sch

02 Minimal independent fail-off and captured ARM context

File: 02_interlocks.kicad_sch

02 Minimal independent fail-off and captured ARM context continued

File: 02_interlocksb1.kicad_sch

03 External protected pack and exclusive conversion populations

File: 03_input_power.kicad_sch

03 External protected pack and exclusive conversion populations continued

File: 03_input_powerb1.kicad_sch

04 2S boost,10.9/13.8V candidate

File: 04_boost.kicad_sch

05 Sine reconstruction and training command

File: 05_pwm_training_command.kicad_sch

07 Independent very-low-power sine-AM training path

File: 07_training.kicad_sch

07 Independent very-low-power sine-AM training path continued

File: 07_trainingb1.kicad_sch

08 Local voltage, current and thermal trip

File: 08_protection.kicad_sch

08 Local voltage, current and thermal trip continued

File: 08_protectionb1.kicad_sch

06 DL3BBX retained72MHz doubler, driver, final and output network

File: 06_dl3_rf.kicad_sch

06b Retained DL3BBX T3 doubler and T4 driver

File: 06b_dl3_doubler_driver.kicad_sch

06c Retained DL3BBX MRF237 final and output matching/filter

File: 06c_dl3_final.kicad_sch

10 Original Darlington collector-supply AM, PWM sine input

File: 10_original_modulator.kicad_sch

11 External supply declarations and qualification boundary

File: 11_external_contract.kicad_sch

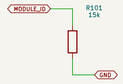
ARDF-TX / NOT RELEASED
Sheet: /
File: RF2M.kicad_sch
Title: ARDF-TX 2 m A2A evaluation schematic
Size: A2 | Date: 2026-10-15 | Rev: M5A-R3
KiCad E.D.A. 10.0.6 | Id: 1/17

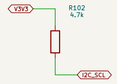
01 Interface and local Si5351 source

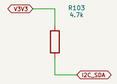
R3: no RF/analog/thermal hardware validation. Original RF attribution and open procurement in MSA_R3 documents.

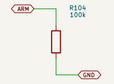
J101
Controller 20 pin

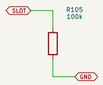


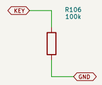


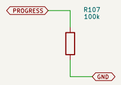


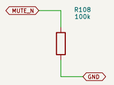


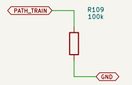


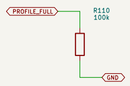


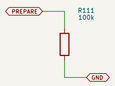


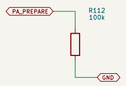


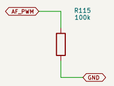


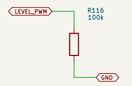




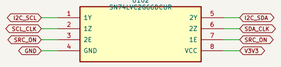


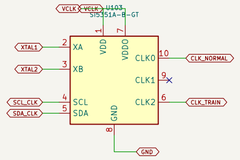


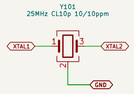


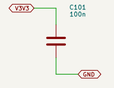


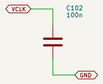


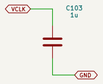
















MSA-R3: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only: no ordering or manufacture

ARDF-TX

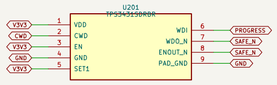
Sheet: /01 Interface and local Si5351 source/
File: 01_Interface_source.kicad_sch

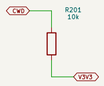
Title: 01 Interface and local Si5351 source

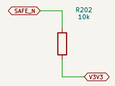
Size: A2	Date: 2026-10-15	**Rev: MSA-R3**
KiCad E.D.A. 10.0.6		Id: 2/17

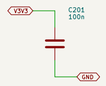
02 Minimal independent fail-off and captured ARM context

R3: no RF/analog/thermal hardware validation. Original RF attribution and open procurement in M5A_R3 documents.

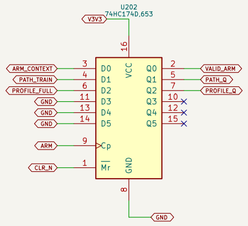














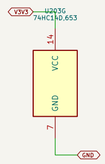


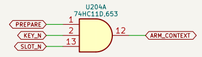




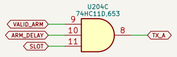


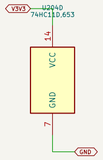


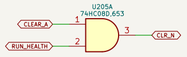


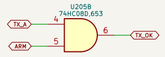


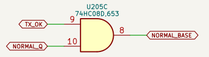


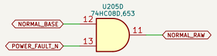


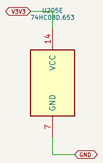


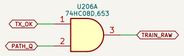


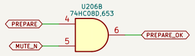


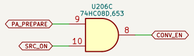


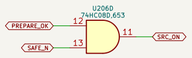


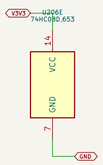


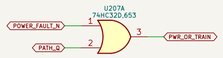




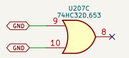












M5A-R3: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only; no ordering or manufacture

ARDF-TX

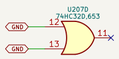
Sheet: /02 Minimal independent fail-off and captured ARM context/
File: 02_interlocks.kicad_sch

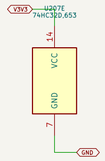
Title: 02 Minimal independent fail-off and captured ARM context

Size: A2	Date: 2026-10-15	**Rev: M5A-R3**
KiCad E.D.A. 10.0.6		Id: 3/17

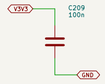
02 Minimal independent fail-off and captured ARM context continued

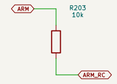
R3: no RF/analog/thermal hardware validation. Original RF attribution and open procurement in M5A_R3 documents.



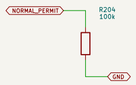


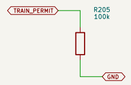


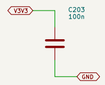


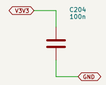


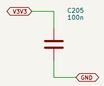


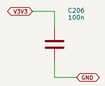


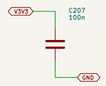


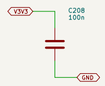


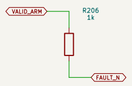
























M5A-R3: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only: no ordering or manufacture

ARDF-TX

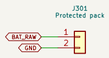
Sheet: /02 Minimal independent fail-off and captured ARM context continued/
File: 02_Interlocksb1.kicad_sch

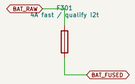
Title: 02 Minimal independent fail-off and captured ARM context continued

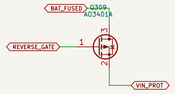
Size: A2	Date: 2026-10-15	**Rev: M5A-R3**
KiCad E.D.A. 10.0.6		Id: 4/17

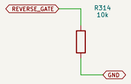
03 External protected pack and exclusive conversion populations

R3: no RF/analog/thermal hardware validation. Original RF attribution and open procurement in M5A_R3 documents.



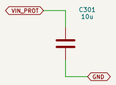


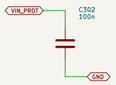


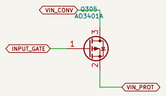


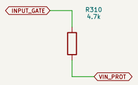


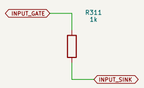


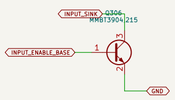


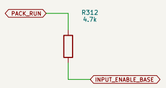


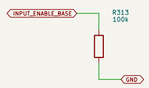




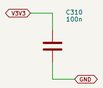




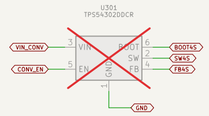


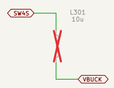


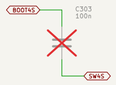


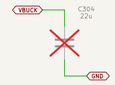


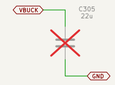






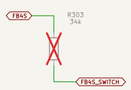


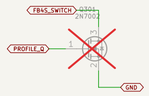


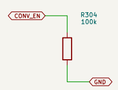


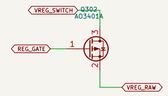


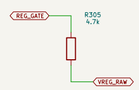












M5A-R3: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only; no ordering or manufacture

ARDF-TX

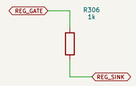
Sheet: /03 External protected pack and exclusive conversion populations/
File: 03_Input_power.kicad_sch

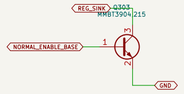
Title: 03 External protected pack and exclusive conversion populations

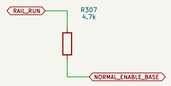
Size: A2	Date: 2026-10-15	**Rev: M5A-R3**
KiCad E.D.A. 10.0.6		Id: 5/17

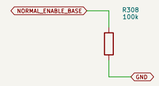
03 External protected pack and exclusive conversion populations continued

R3: no RF/analog/thermal hardware validation. Original RF attribution and open procurement in M5A_R3 documents.

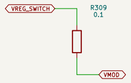






















M5A-R3: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only: no ordering or manufacture

ARDF-TX

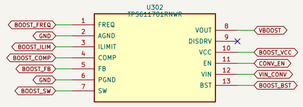
Sheet: /03 External protected pack and exclusive conversion populations continued/
File: 03_input_powerb1.kicad_sch

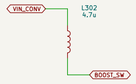
Title: 03 External protected pack and exclusive conversion populations continued

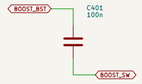
Size: A2	Date: 2026-10-15	**Rev: M5A-R3**
KiCad E.D.A. 10.0.6		Id: 6/17

04 2S boost,10.9/13.8V candidate

R3: no RF/analog/thermal hardware validation. Original RF attribution and open procurement in MSA_R3 documents.



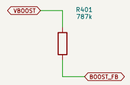


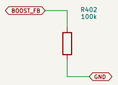


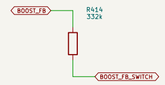


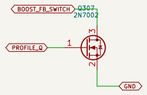


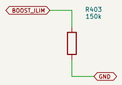


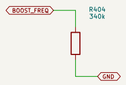


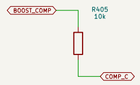


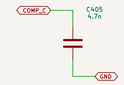


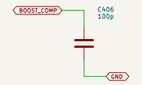












MSA-R3: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only: no ordering or manufacture

ARDF-TX

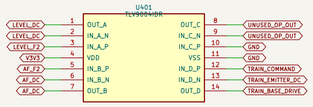
Sheet: /04 2S boost,10.9/13.8V candidate/
File: 04_boost.kicad_sch

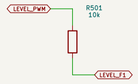
Title: 04 2S boost,10.9/13.8V candidate

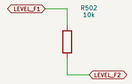
Size: A2	Date: 2026-10-15	**Rev: MSA-R3**
KiCad E.D.A. 10.0.6		Id: 7/17

05 Sine reconstruction and training command

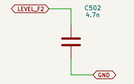
R3: no RF/analog/thermal hardware validation. Original RF attribution and open procurement in MSA_R3 documents.

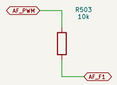


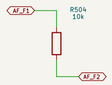




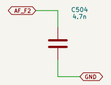


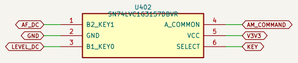


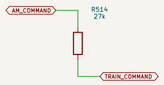


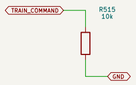


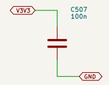


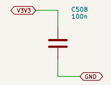












MSA-R3: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only: no ordering or manufacture

ARDF-TX

Sheet: /05 Sine reconstruction and training command/
File: 05_pwm_training_command.kicad_sch

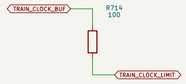
Title: 05 Sine reconstruction and training command

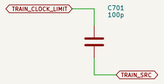
Size: A2	Date: 2026-10-15	**Rev: MSA-R3**
KiCad E.D.A. 10.0.6		Id: 8/17

07 Independent very-low-power sine-AM training path

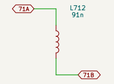
R3: no RF/analog/thermal hardware validation. Original RF attribution and open procurement in M5A_R3 documents.

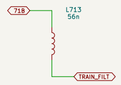


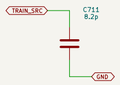


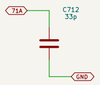


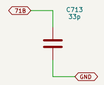




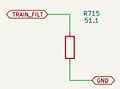


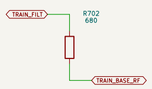


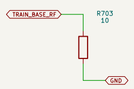


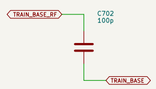


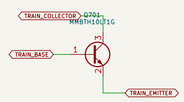


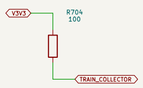


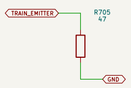


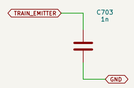


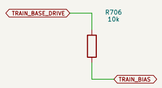


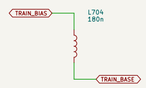


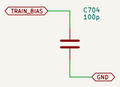


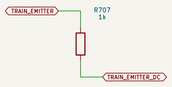


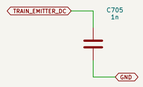


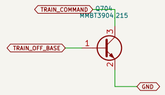


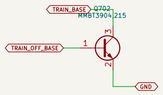


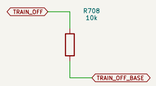


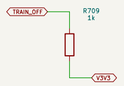


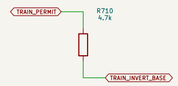


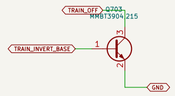














M5A-R3: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only: no ordering or manufacture

ARDF-TX

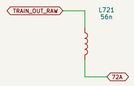
Sheet: /07 Independent very-low-power sine-AM training path/
File: 07_training.kicad_sch

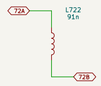
Title: 07 Independent very-low-power sine-AM training path

Size: A2	Date: 2026-10-15	**Rev: M5A-R3**
KiCad E.D.A. 10.0.6		Id: 9/17

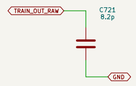
07 Independent very-low-power sine-AM training path continued

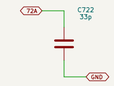
R3: no RF/analog/thermal hardware validation. Original RF attribution and open procurement in M5A_R3 documents.

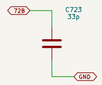


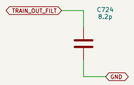


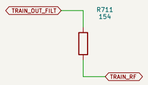


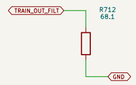


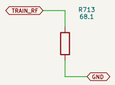


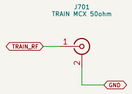
















M5A-R3: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only: no ordering or manufacture

ARDF-TX

Sheet: /07 Independent very-low-power sine-AM training path continued/
File: 07_trainingb1.kicad_sch

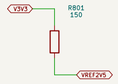
Title: 07 Independent very-low-power sine-AM training path continued

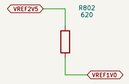
Size: A2	Date: 2026-10-15	**Rev: M5A-R3**
KiCad E.D.A. 10.0.6		Id: 10/17

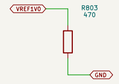
08 Local voltage, current and thermal trip

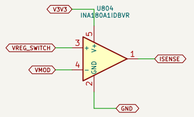
R3: no RF/analog/thermal hardware validation. Original RF attribution and open procurement in MSA_R3 documents.

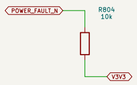


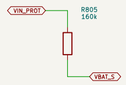


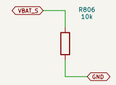


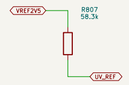


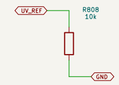


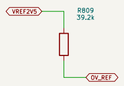


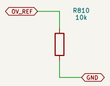


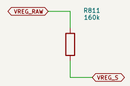


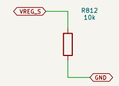


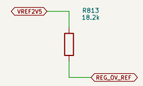


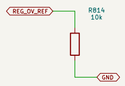


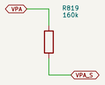


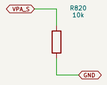


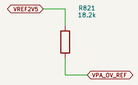


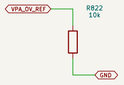


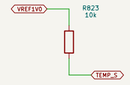


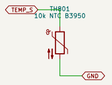


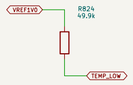


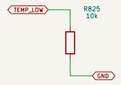


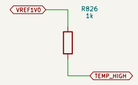


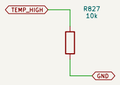












MSA-R3: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only; no ordering or manufacture

ARDF-TX

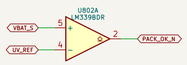
Sheet: /08 Local voltage, current and thermal trip/
File: 08_protection.kicad_sch

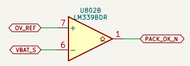
Title: 08 Local voltage, current and thermal trip

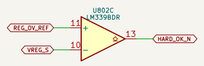
Size: A2	Date: 2026-10-15	**Rev: MSA-R3**
KiCad E.D.A. 10.0.6		Id: 11/17

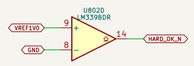
08 Local voltage, current and thermal trip continued

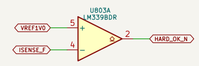
R3: no RF/analog/thermal hardware validation. Original RF attribution and open procurement in M5A_R3 documents.

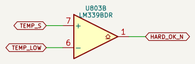


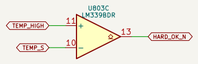


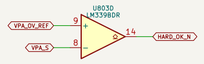


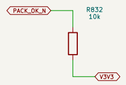


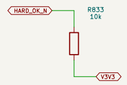








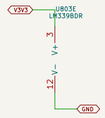


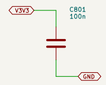


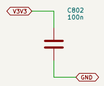


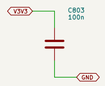


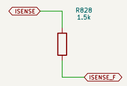


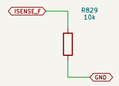


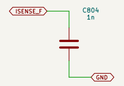






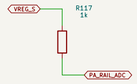


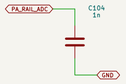


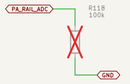












M5A-R3: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only; no ordering or manufacture

ARDF-TX

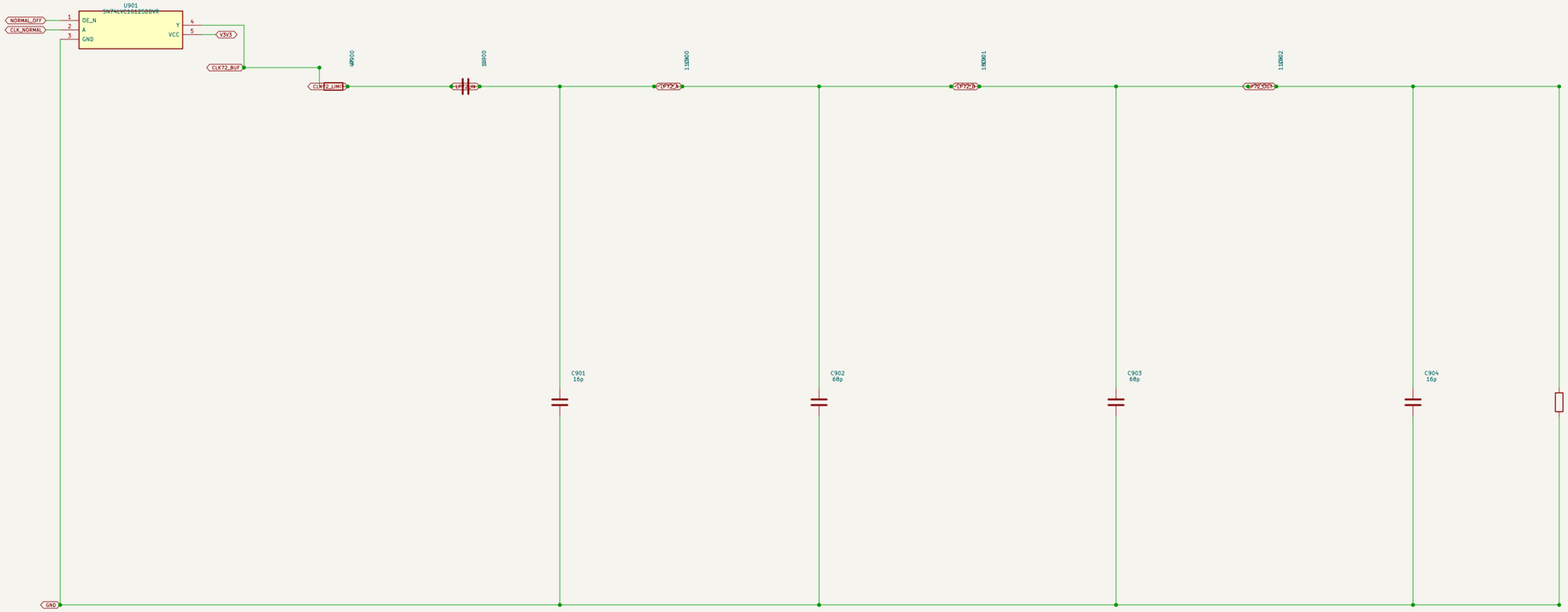
Sheet: /08 Local voltage, current and thermal trip continued/
File: 08_protectionb1.kicad_sch

Title: 08 Local voltage, current and thermal trip continued

Size: A2	Date: 2026-10-15	**Rev: M5A-R3**
KiCad E.D.A. 10.0.6		Id: 12/17

06 DL3BBX retained72MHz doubler, driver, final and output network

Actual connected RF network. Pin numbers and named supply/interface crossings are explicit.
DL3BBX CQ-DL5/1992 reference: PRIVATE CONSTRUCTION ONLY. Derivative RF performance NOT_RUN.



OPEN: exact RF transistor supplier/manufacturer and landings; actual source drive, tuning, sineAM power, spectrum and stability.
Do not infer validated performance or manufacture readiness from the original circuit or a passing ERC.

M5A-R3: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only; no ordering or manufacture

ARDF-TX

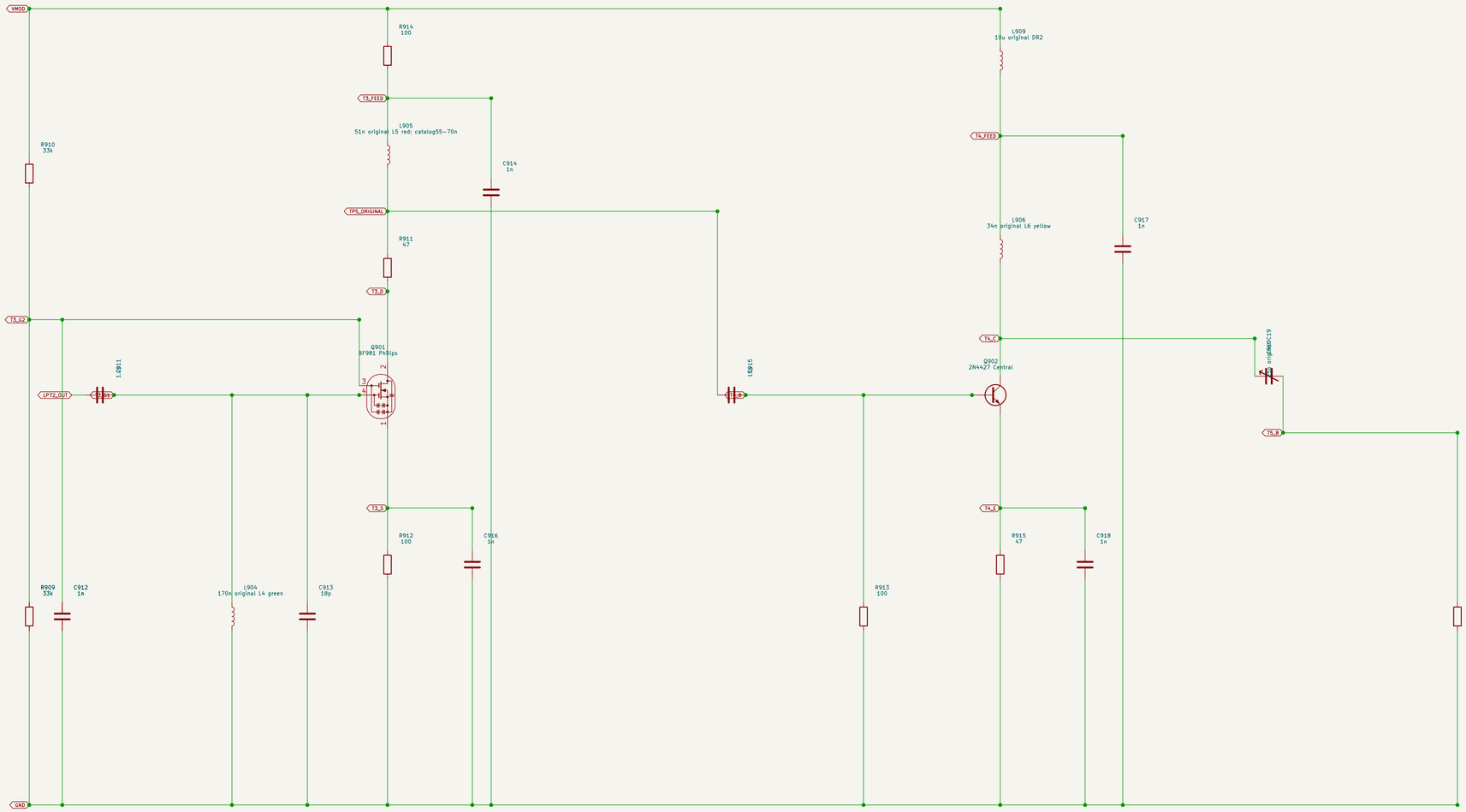
Sheet: /06 DL3BBX retained72MHz doubler, driver, final and output network/
File: 06_dl3_rf.kicad_sch

Title: 06 DL3BBX retained72MHz doubler, driver, final and output network

Size: A2	Date: 2026-10-15	**Rev: M5A-R3**
KiCad E.D.A. 10.0.6		Id: 13/17

06b Retained DL3BBX T3 doubler and T4 driver

Actual connected RF network. Pin numbers and named supply/interface crossings are explicit.
DL3BBX CQ-DL5/1992 reference: PRIVATE CONSTRUCTION ONLY. Derivative RF performance NOT_RUN.



OPEN: exact RF transistor supplier/manufacturer and landings; actual source drive, tuning, sineAM power, spectrum and stability.
Do not infer validated performance or manufacture readiness from the original circuit or a passing ERC.

M5A-R3: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only: no ordering or manufacture

ARDF-TX

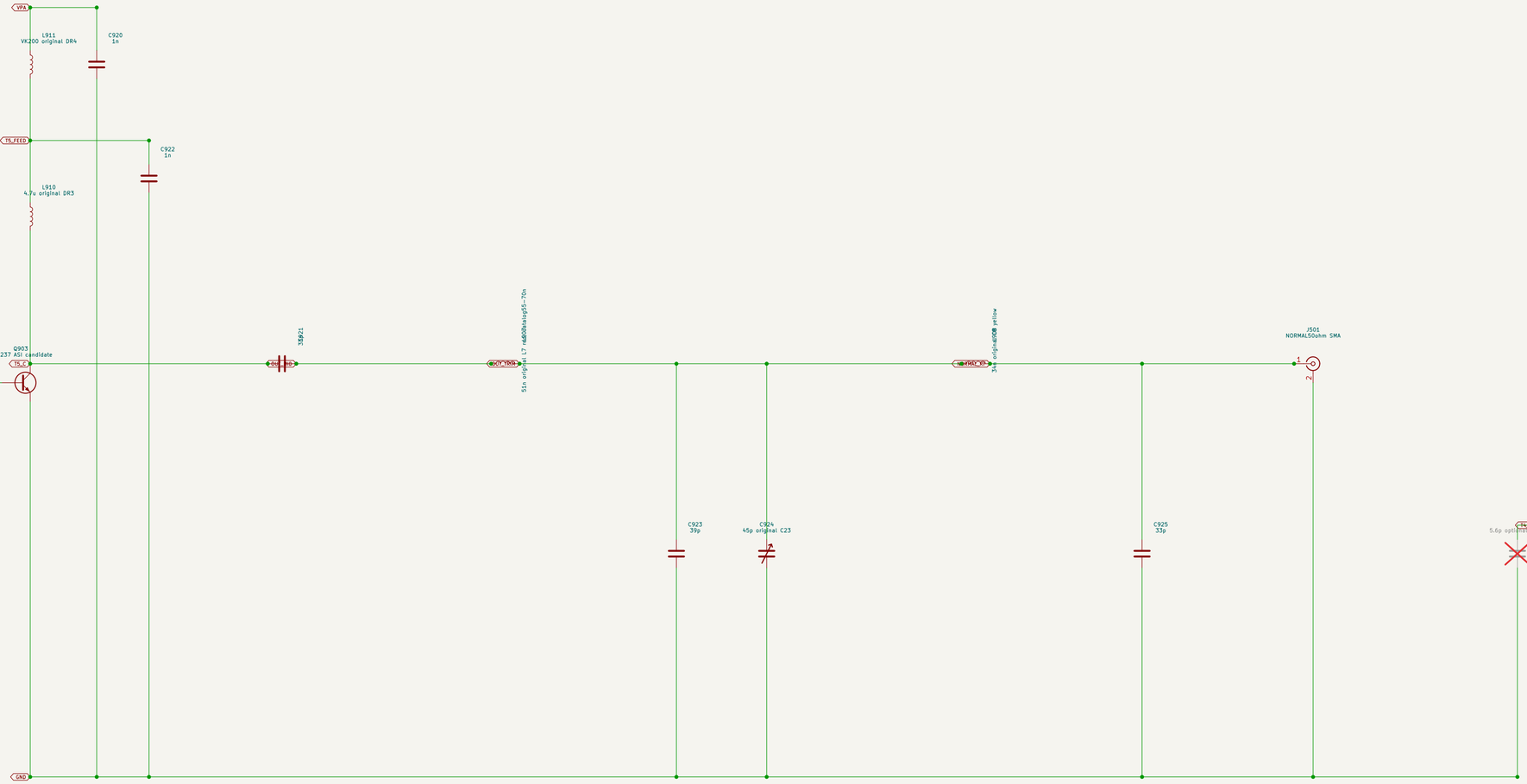
Sheet: /06b Retained DL3BBX T3 doubler and T4 driver/
File: 06b_dl3_doubler_driver.kicad_sch

Title: 06b Retained DL3BBX T3 doubler and T4 driver

Size: A2	Date: 2026-10-15	**Rev: M5A-R3**
KiCad E.D.A. 10.0.6		Id: 14/17

06c Retained DL3BBX MRF237 final and output matching/filter

Actual connected RF network. Pin numbers and named supply/interface crossings are explicit.
DL3BBX CQ-DL5/1992 reference: PRIVATE CONSTRUCTION ONLY. Derivative RF performance NOT_RUN.



OPEN: exact RF transistor supplier/manufacturer and landings; actual source drive, tuning, sineAM power, spectrum and stability.
Do not infer validated performance or manufacture readiness from the original circuit or a passing ERC.

M5A-R3: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

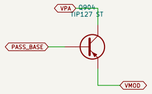
Evaluation review only; no ordering or manufacture
AR0F-TX
Sheet: /06c Retained DL3BBX MRF237 final and output matching/filter/
File: 06c_dl3_final.kicad_sch

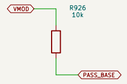
Title: 06c Retained DL3BBX MRF237 final and output matching/filter

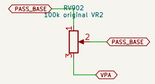
Size: A2	Date: 2026-10-15	**Rev: M5A-R3**
KiCad E.D.A. 10.0.6		Id: 15/17

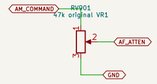
10 Original Darlington collector-supply AM, PWM sine input

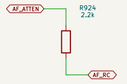
Original T8/R24-26/VR1/VR2/C26/C27 mechanism retained: ST TIP127 replaces obsolete BD676 with documented different package/pins.
No new RF final, premium MMIC or high-power op-amp servo. Profile setting changes converter rail, not live RF path. RF75% AM remains NOT_MEASURED.



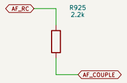


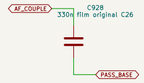


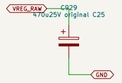




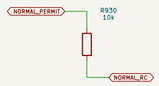


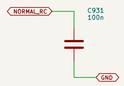


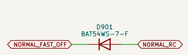


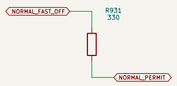


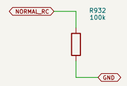


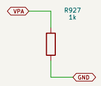












M5A-R3: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only; no ordering or manufacture

ARDF-TX

Sheet: /10 Original Darlington collector-supply AM, PWM sine input/
File: 10_original_modulator.kicad_sch

Title: 10 Original Darlington collector-supply AM, PWM sine input

Size: A2	Date: 2026-10-15	**Rev: M5A-R3**
KiCad E.D.A. 10.0.6		Id: 16/17

11 External supply declarations and qualification boundary

No PCB is released: exact final manufacturer/lot, some magnetic/choke/trimmer geometry and RF source coupling remain OPEN.
No physical measurements exist. See M5A_R3_REVIEW_DECISION.md. PWR_FLAG is an ERC declaration, not a safety approval.



















M5A-R3: REVIEW / ALL hardware measurements NOT RUN / no ordering or manufacture.

Evaluation review only: no ordering or manufacture

ARDF-TX

Sheet: /11 External supply declarations and qualification boundary/
File: 11_external_contract.kicad_sch

Title: 11 External supply declarations and qualification boundary

Size: A2	Date: 2026-10-15	**Rev: M5A-R3**
KiCad E.D.A. 10.0.6		Id: 17/17